

RichAuto A18-4Axis Operation Manual

Electrical connection

Circuit

Start the machine

Common key functions

Common operation

Return to mechanical home

Workpiece origin setting

manually set XYZ

Automatically set Z

Workpiece origin setting (3AXIS)

XY coordinate settings

Z coordinate settings

Workpiece origin setting (4AXIS)-C Axis

XY coordinate settings

Z coordinate settings

A coordinate settings

Switch working coordinates or machine coordinates

Return to workpiece origin

Set the manual moving speed

Operating spindle

Run the processing program

suspend or cancel processing

Emergency stop

Backup and restoration

backup parameters

Recovery parameters

Sample program

A11 User Manual

Maintenance and maintenance

Spindle part

Water-cooled spindle

Air-cooled spindle

Lubrication part



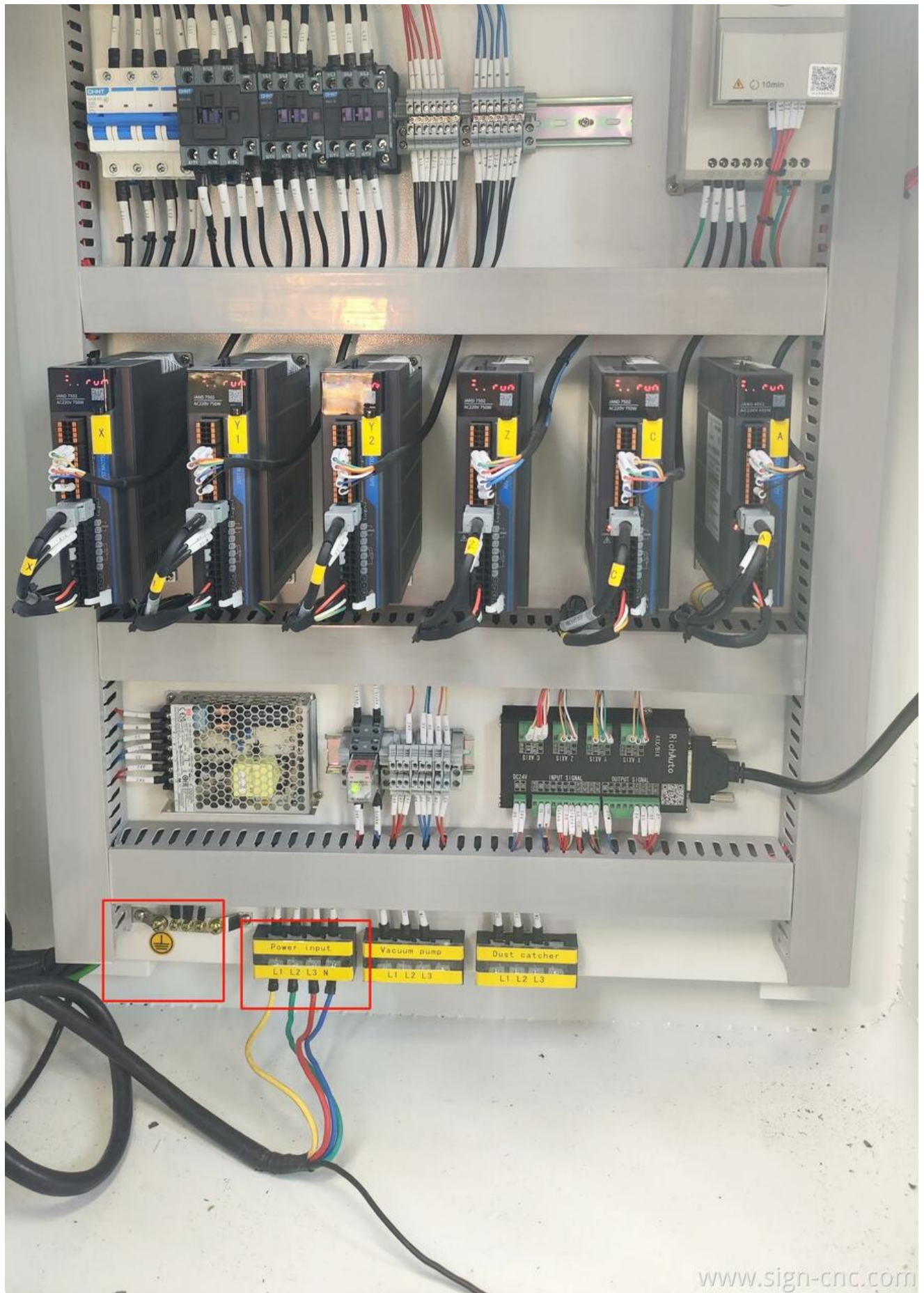


Electrical connection

Circuit

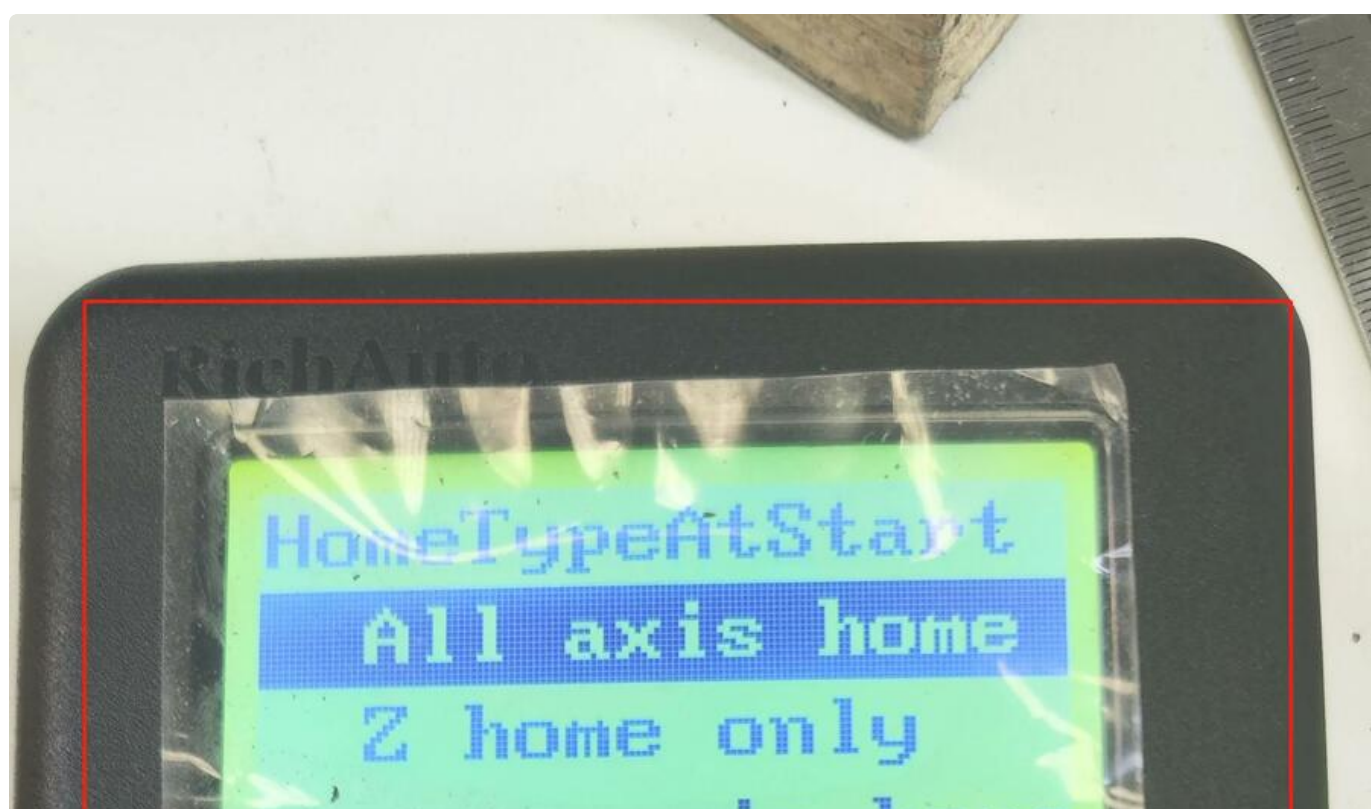
- connect the power supply conforming to the machine. When the incoming line is 380V, pay attention to the position of the zero line.
- During long-term use, the grounding terminal of the chassis must be connected with the ground wire.
- Closed circuit breaker main switch



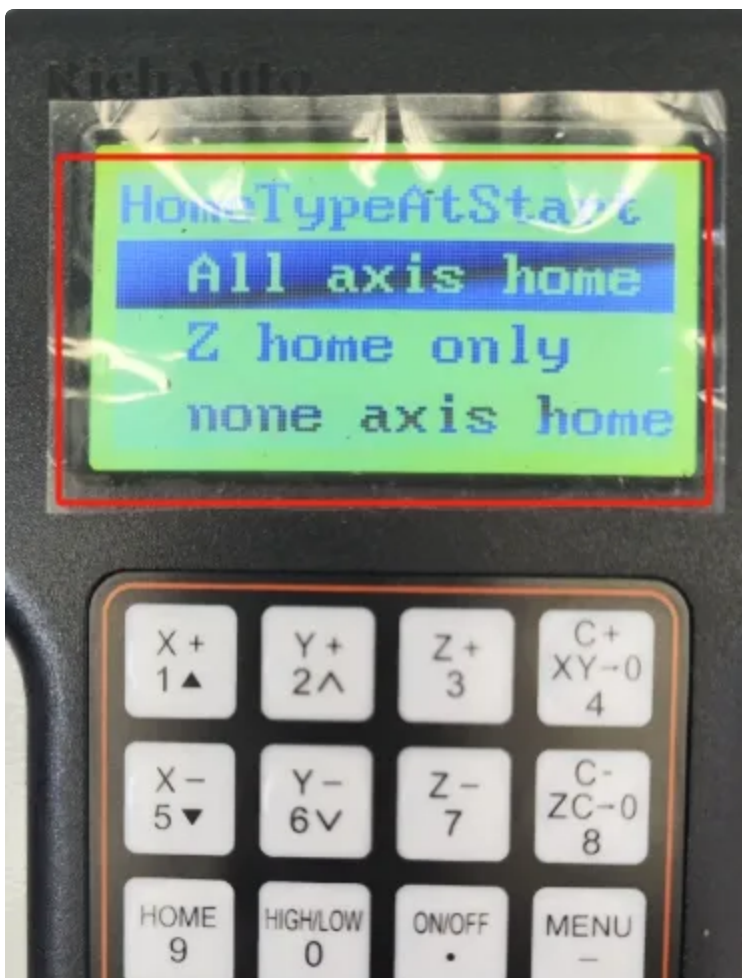


Start the machine

turn on the switch. When the green light is on, the power supply of the machine is normal.
When the following interface is displayed on the handle, the machine starts normally.









3.WARNING: BEFORE EVERY PROCESSING DOCUMENT, MAKE SURE THAT A- AXIS POSITION IS IN MECHANICAL COORDINATES "0"

Common key functions

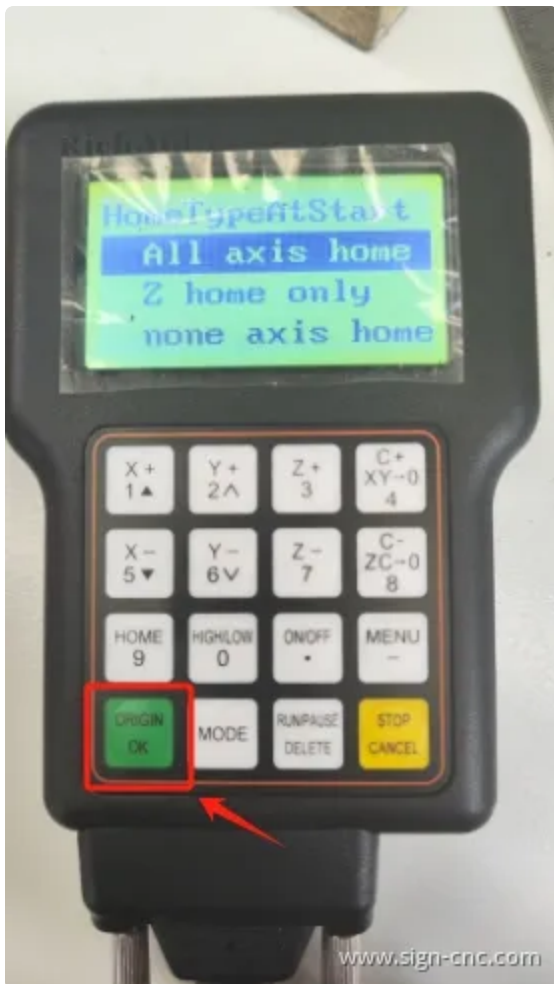
key	function	remarks
X+	move the x axis in the positive direction	in Continue mode
X-	move the x axis to the negative direction	in Continue mode
Y+	move the y axis in the positive direction	in Continue mode
Y-	move the y axis to the negative direction	in Continue mode
Z+	move in the positive direction Z axis	in Continue mode
Z-	move to negative direction Z axis	in Continue mode
C+	move in the positive direction C axis	in Continue mode
C-	move to negative direction C axis	in Continue mode

ON/OFF+XY→0	XY0 point for workpiece coordinate system setting	workpiece coordinate system mode
ON/OFF+ZC→0	workpiece coordinate system setting Z0/C0 point	
HOME	Back to the mechanical origin	
HIGH/LOW	high-speed mobile and low-speed mobile switching	
ON/OFF	spindle start /stop	
MENU	menu entry button	
ORIGIN OK	return to workpiece origin	
MODE	select mode	continuous, step, and length
DELETE RUN/PAUSE	run delete	
STOP CANCEL	stop program running cancel or return	
MENU +0	enter the mechanical coordinate system	
MENU +1,2,3	select a machining coordinate system	generally, use 1X
MENU + ON/OFF	use tool block automatic cleaning of z-axis height of workpiece coordinates	
ON/OFF + Z+/Z-	select spindle gear	
RUN + HIGH/LOW	advanced Processing	

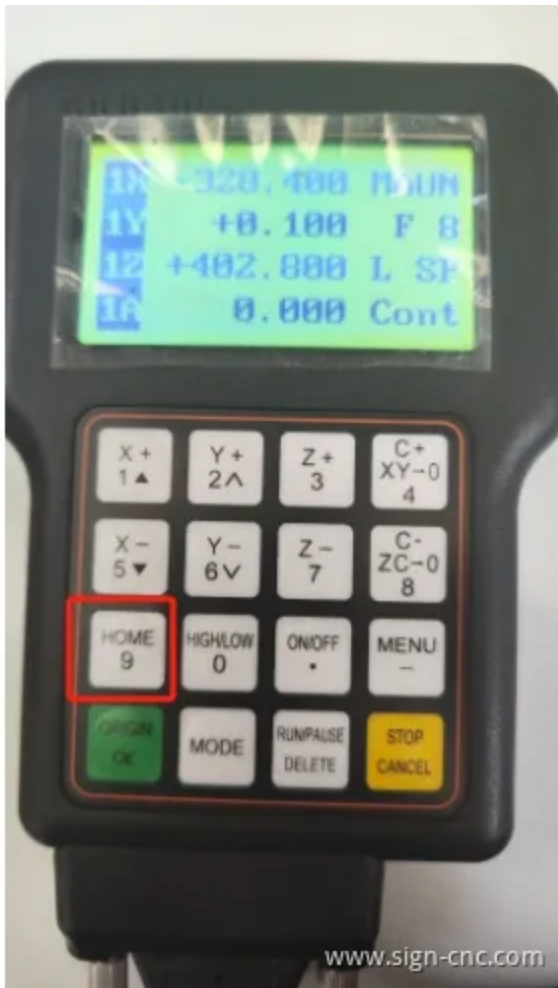
RUN +1,2,3	breakpoint continue	

Common operation

Return to mechanical home



2. In standby mode, as shown in the following figure, click "HOME"



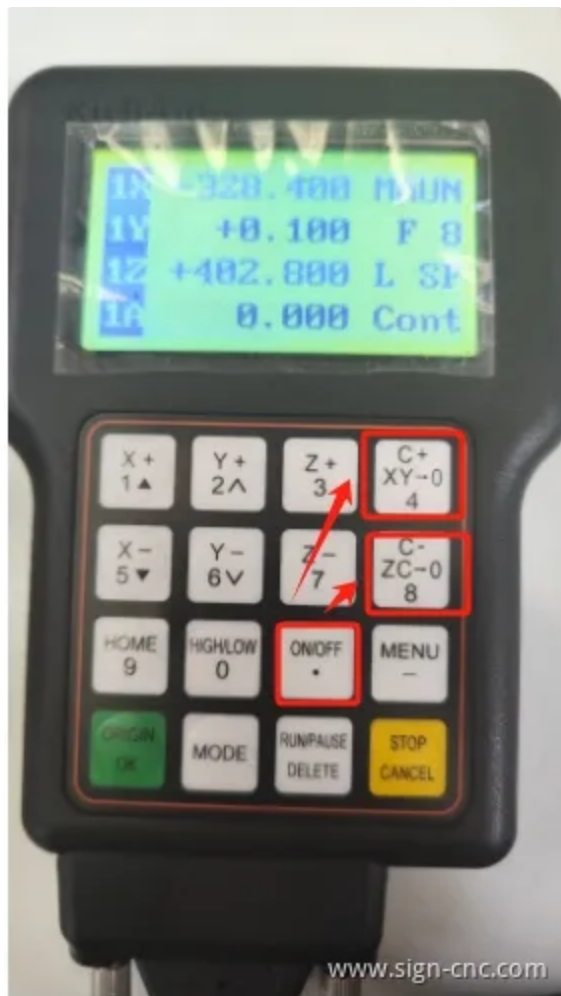
Workpiece origin setting

manually set XYZ

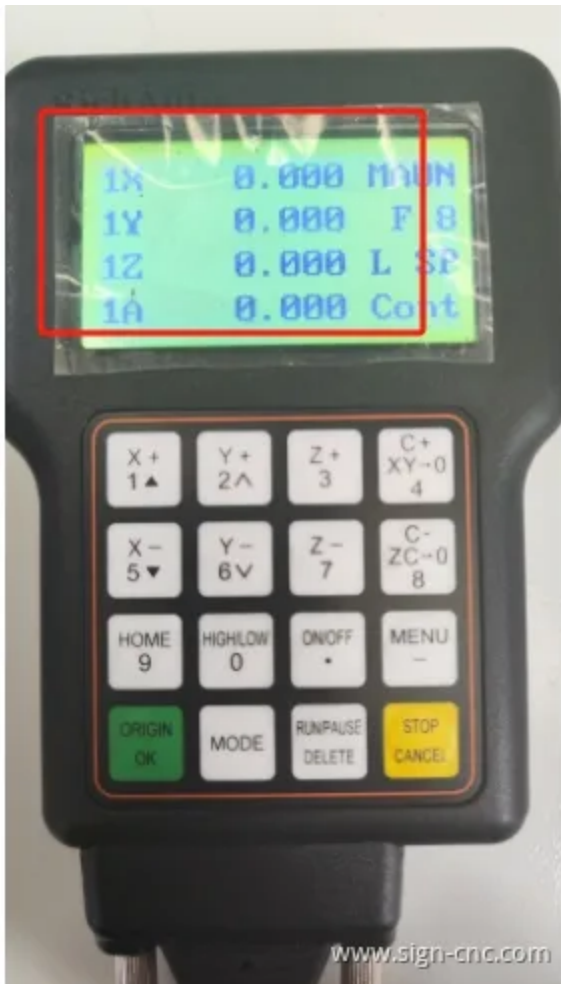
1. Move the machine to the corresponding workpiece origin.



2. As shown in the following figure:



as shown in the following figure, the origin of the workpiece is set.



Automatically set Z

1. As shown in the figure, place the cutter block on the surface of the workpiece.



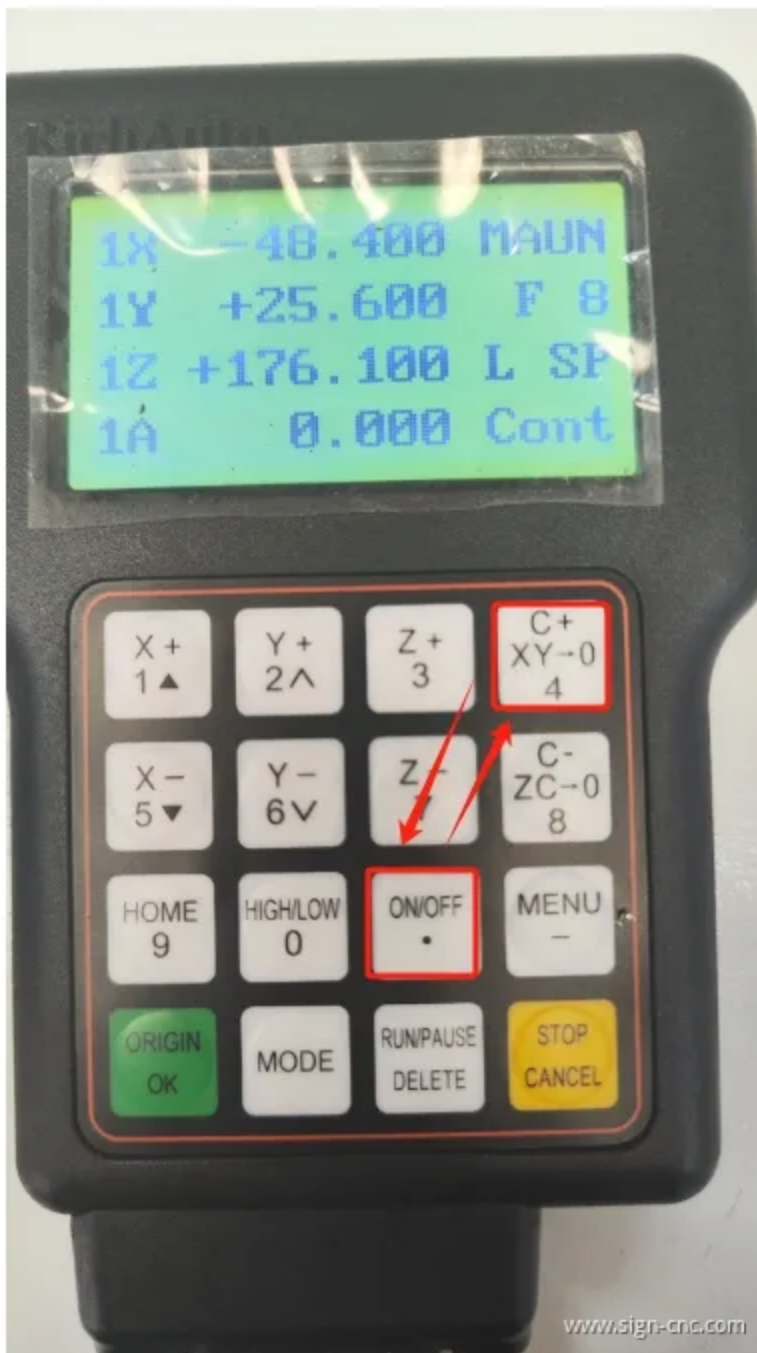
2. Click "ON/OFF" + "MENU" at the same time to set the Z zero coordinate of the workpiece.

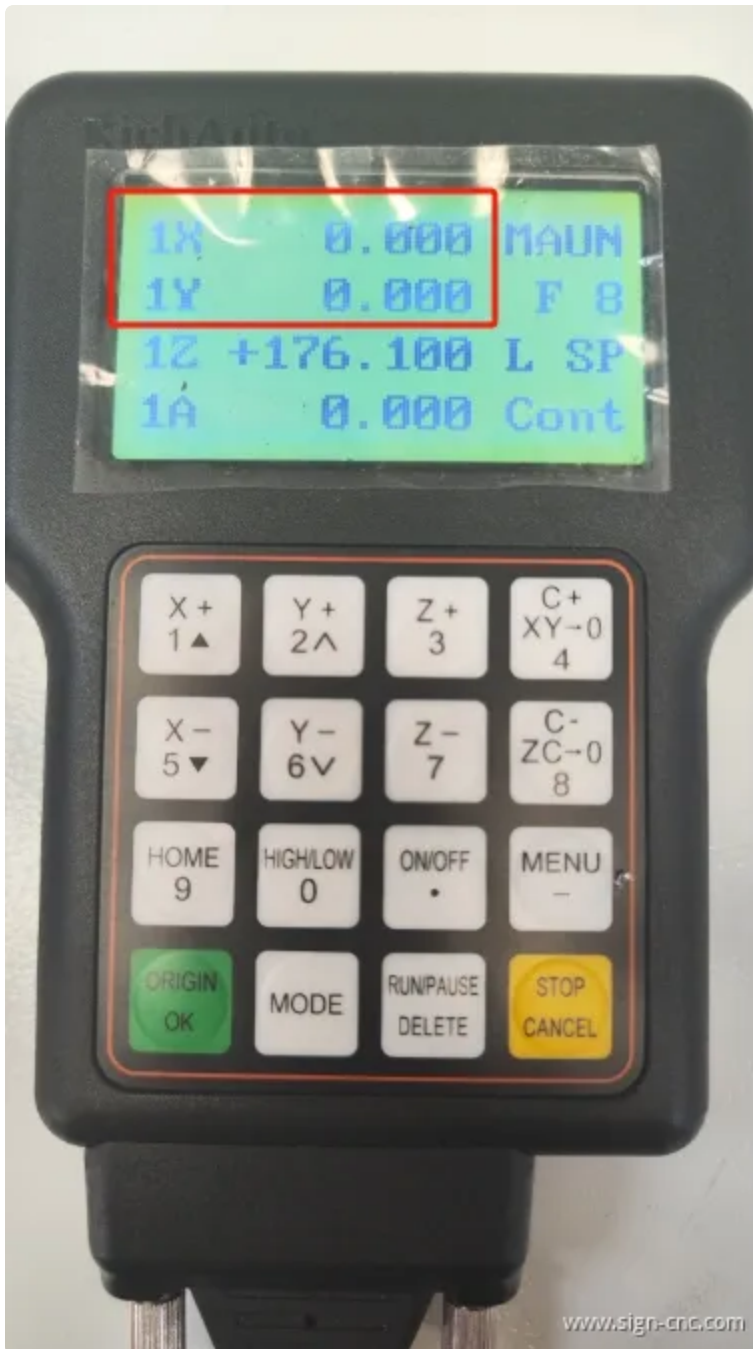
Workpiece origin setting (3AXIS)

XY coordinate settings

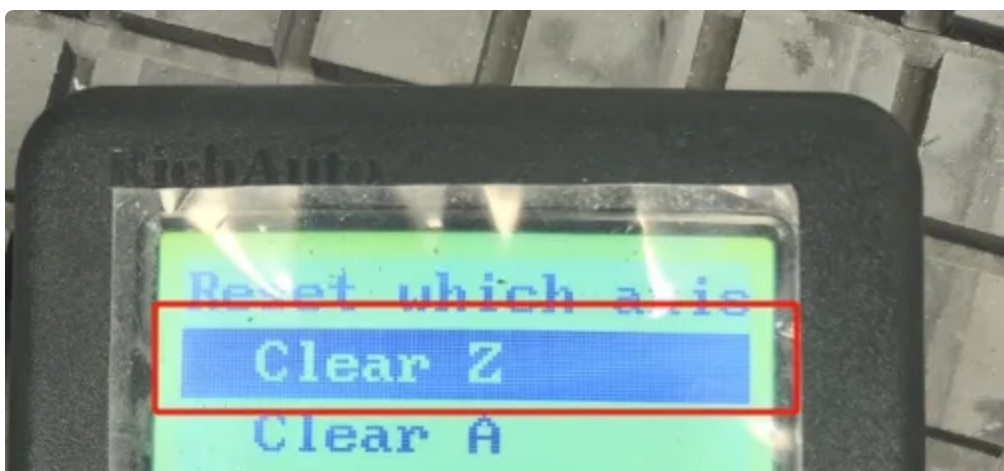


www.sign-cnc.com





Z coordinate settings



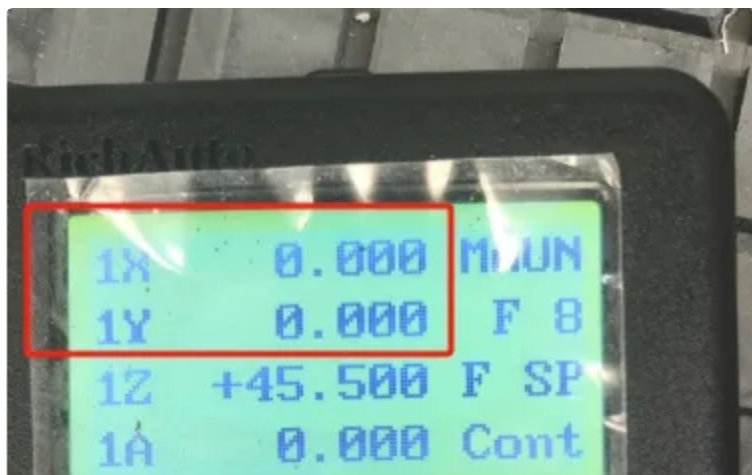
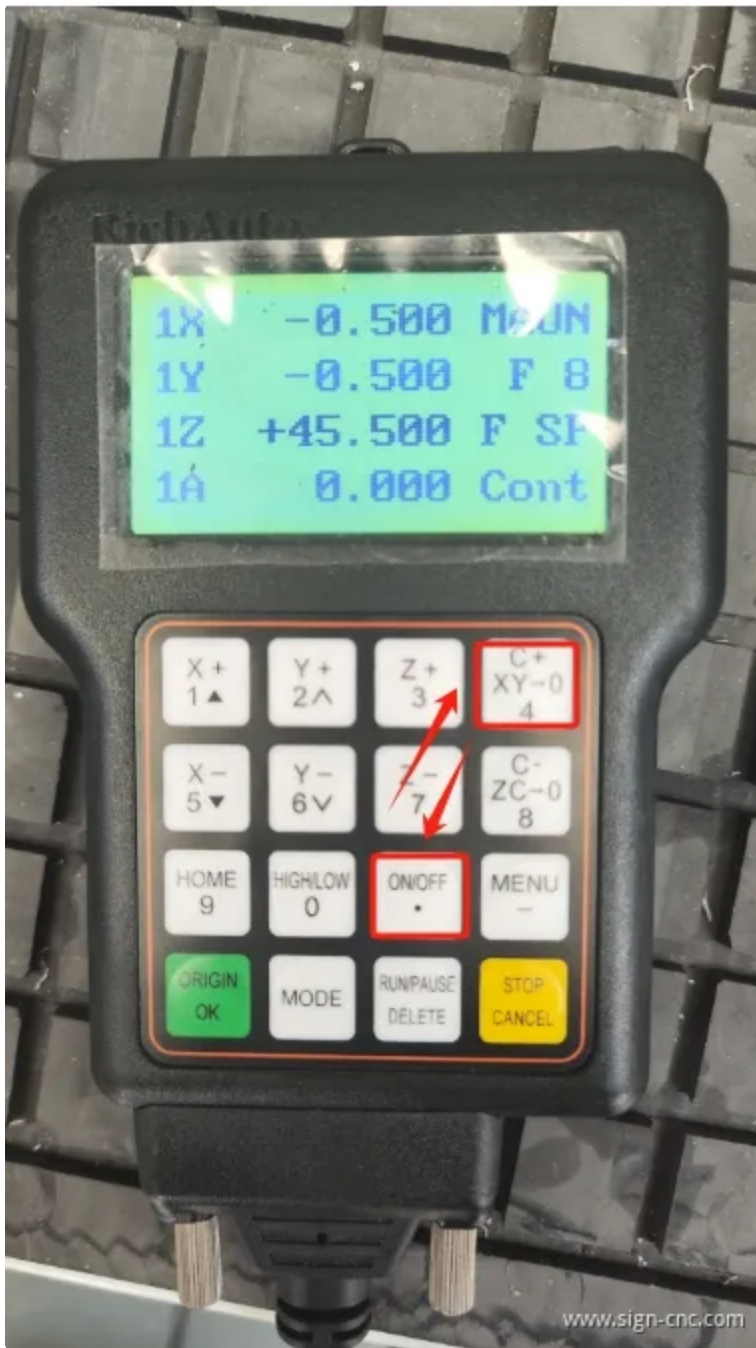




Workpiece origin setting (4AXIS)–C Axis

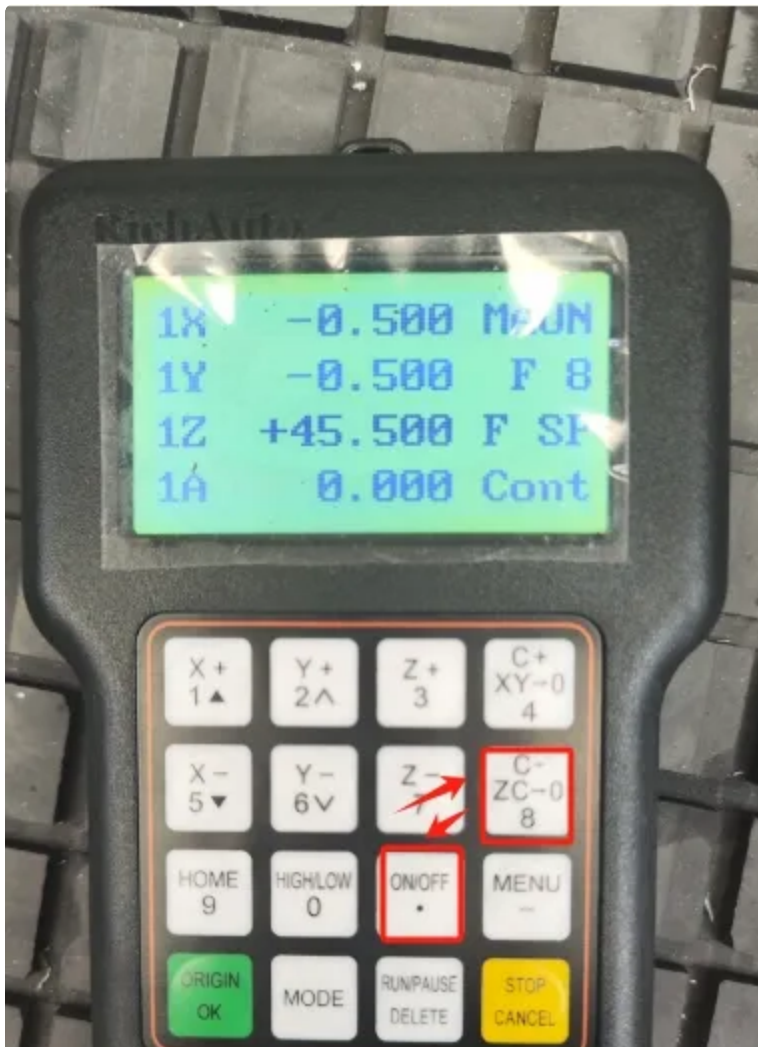
XY coordinate settings

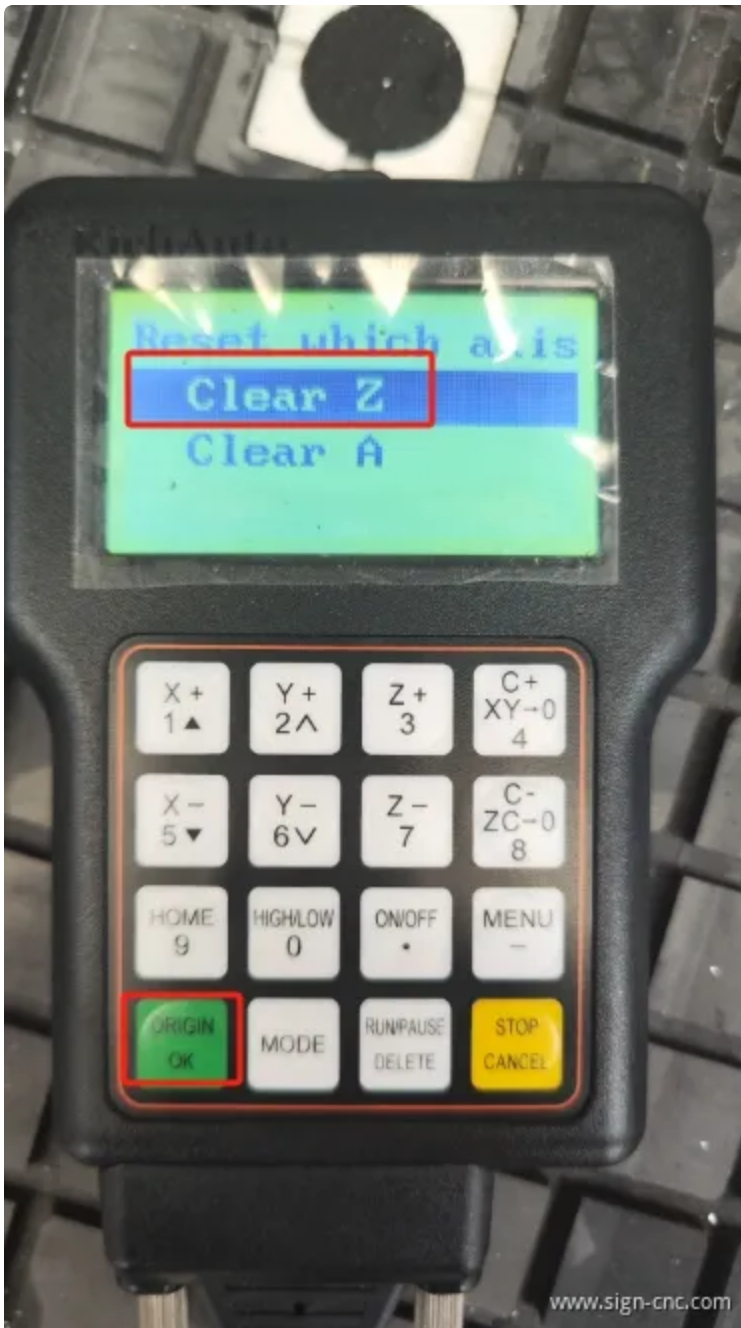






Z coordinate settings





A coordinate settings

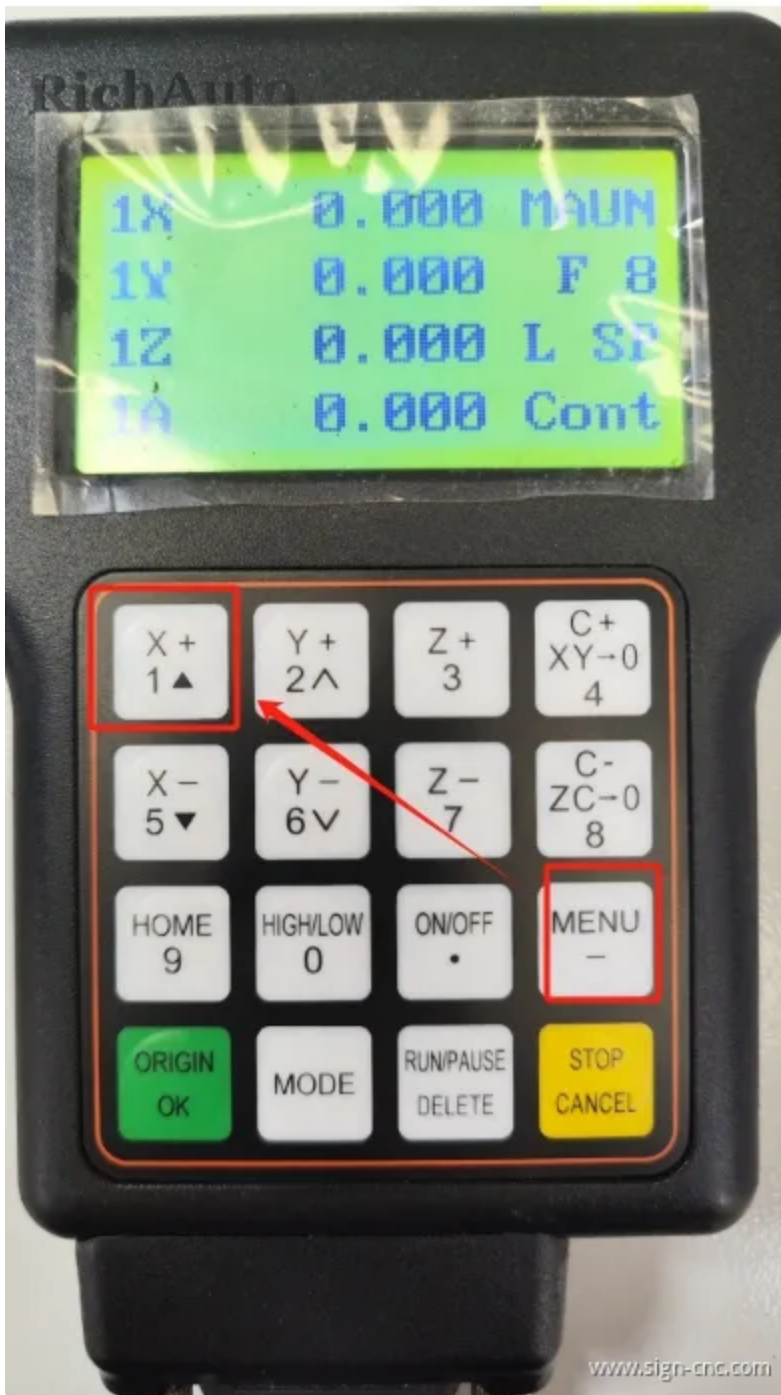
like Z coordinate set

Switch working coordinates or mechine coordinates

Press "MENU" + "0" select mechine coordinates .



Press "MENU" + "1" select NO.1 mechine coordinates . select NO.2 ,press "MENU" + "2".



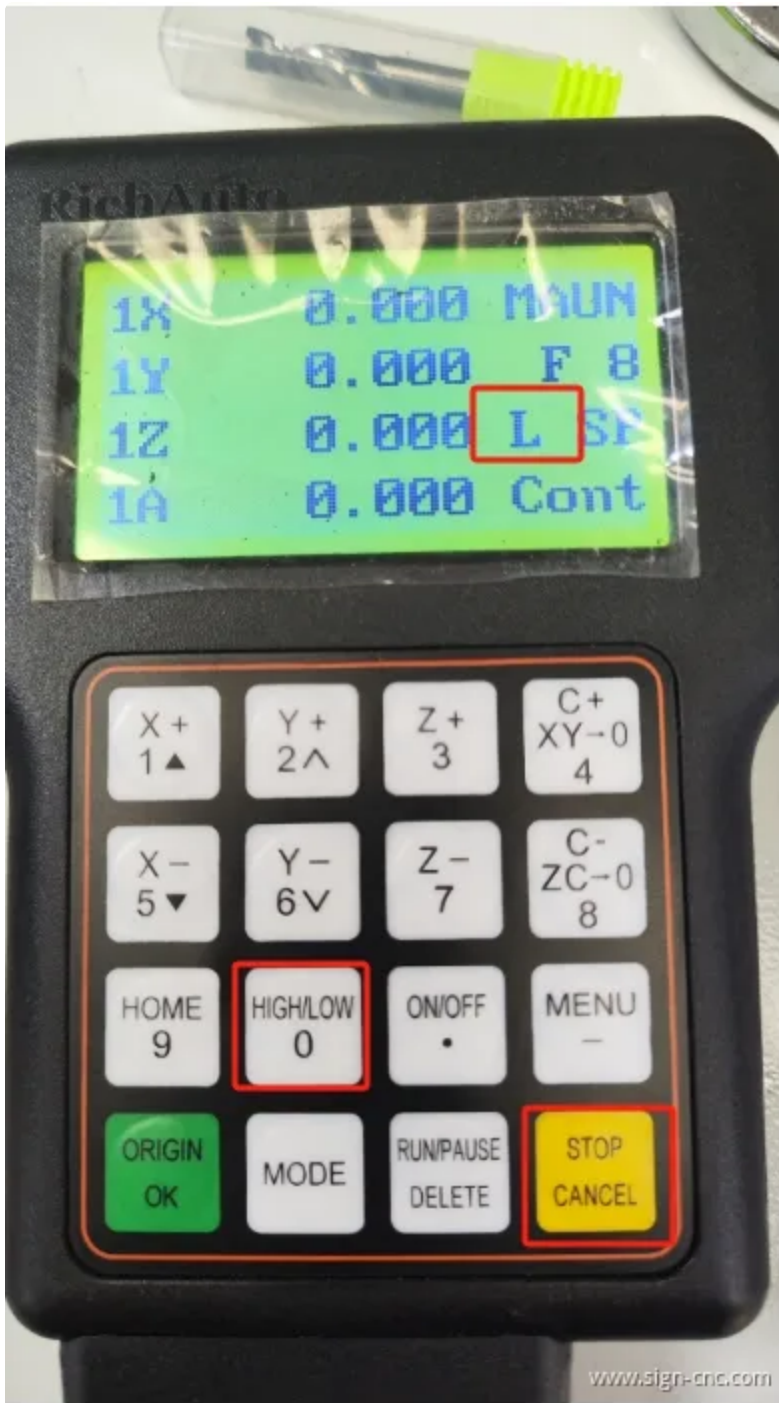
Return to workpiece origin

press the "ORIGIN/OK" button in standby mode, and the machine can return to the ORIGIN of workpiece coordinates.

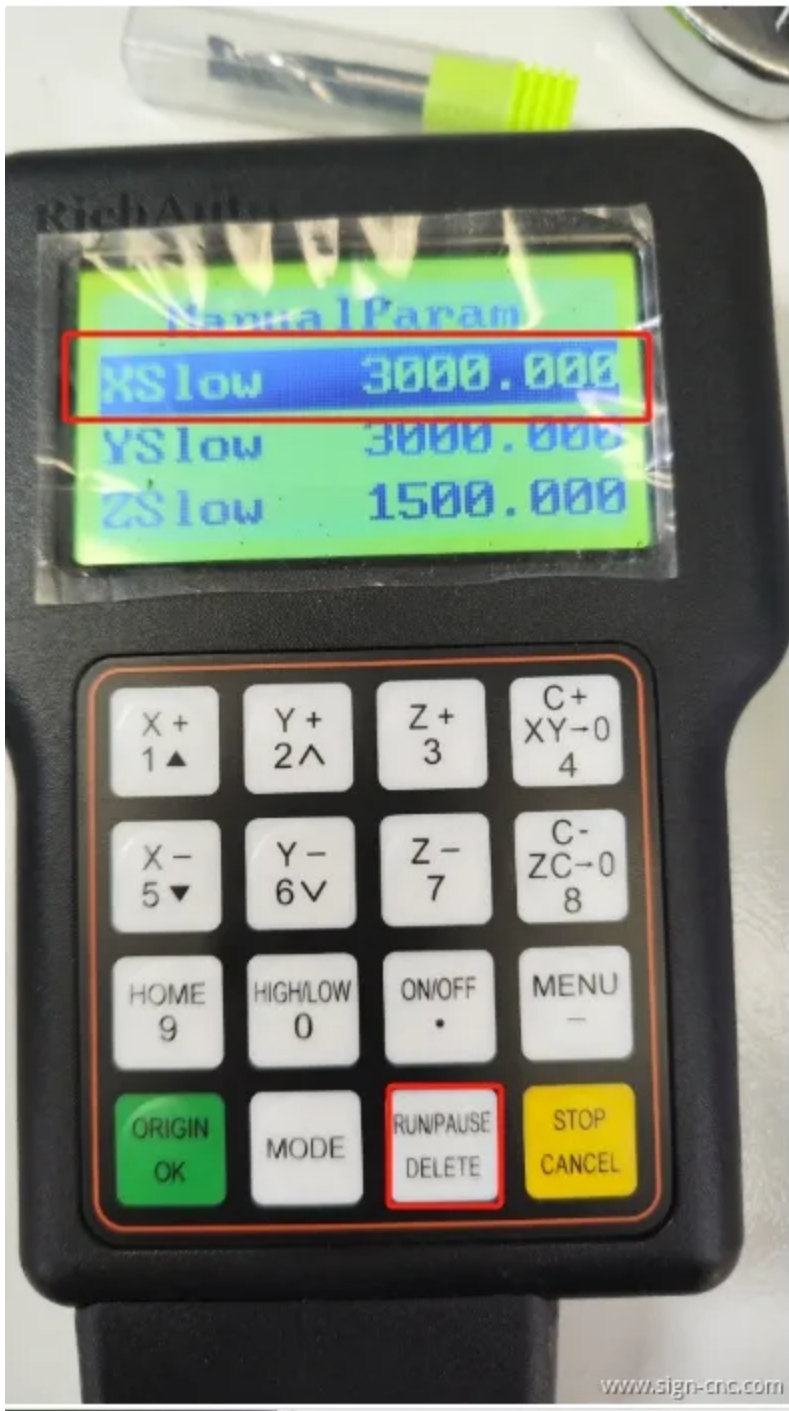


Set the manual moving speed

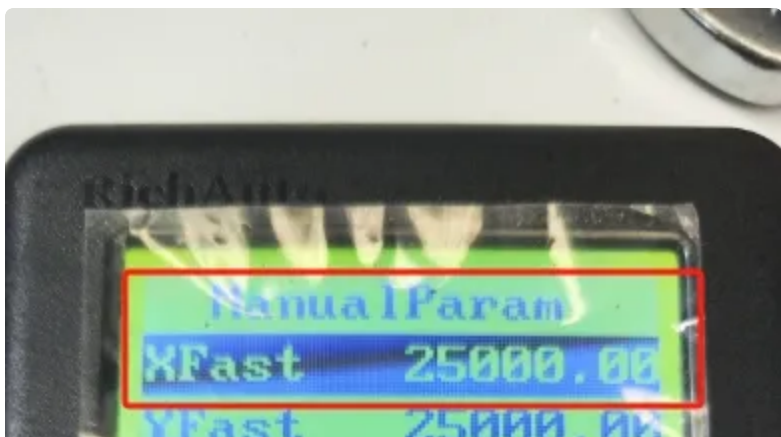
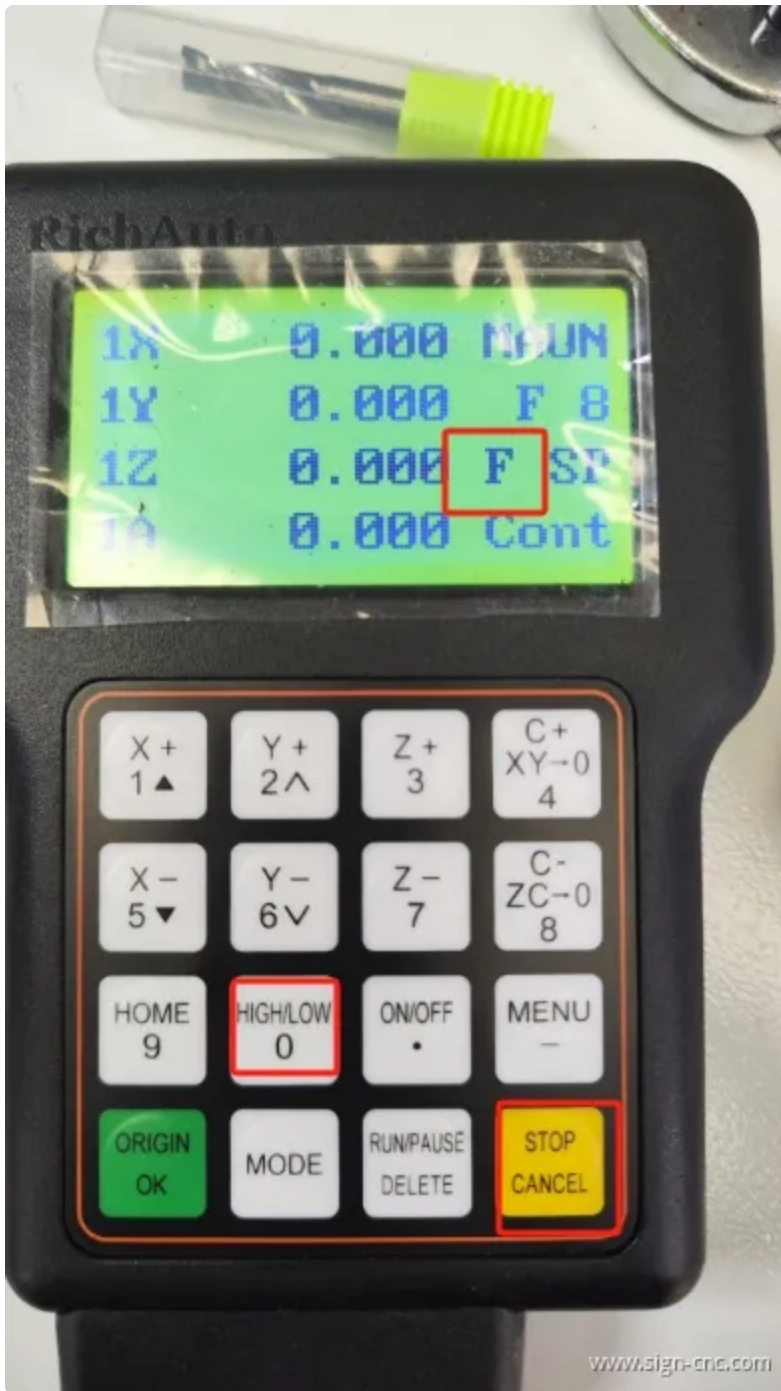
1. As shown in the following figure, under the standby interface, click the "CANCEL" button



2. Enter the slow moving speed setting, select "XSlow" and press the "DELETE" button to set it. Unit: mm/min



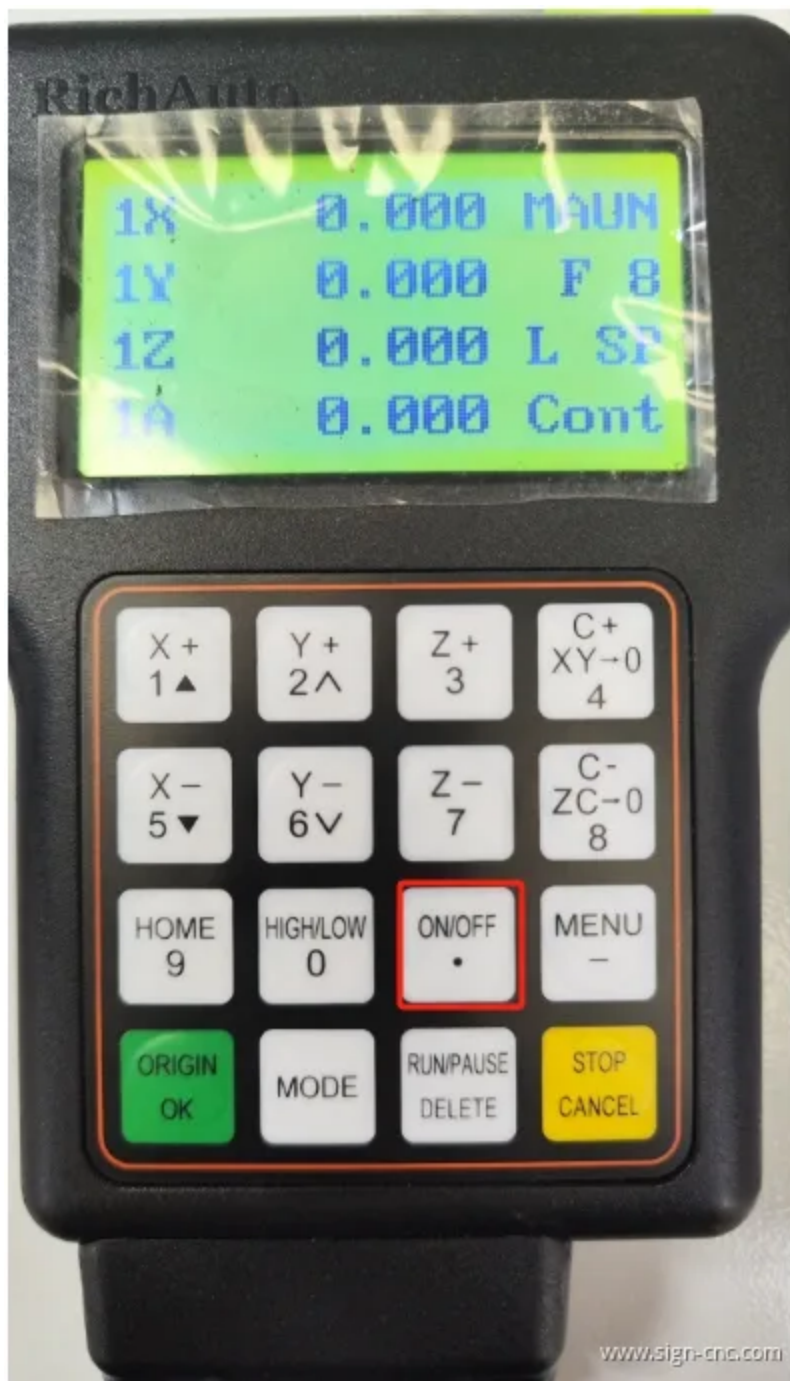
3. Manual fast moving speed can be set in high-speed mode.



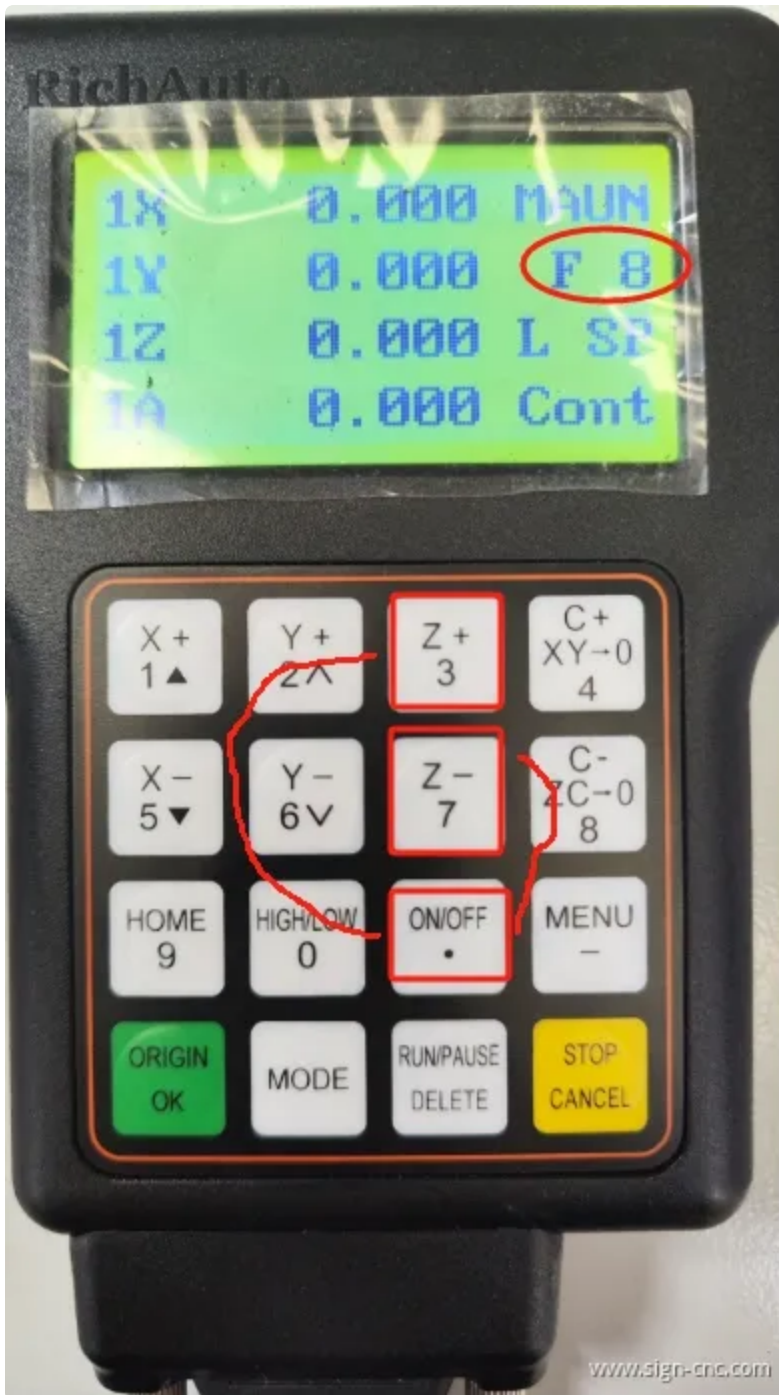


Operating spindle

1、 Press "ON/OFF" ,start or stop the spindle .



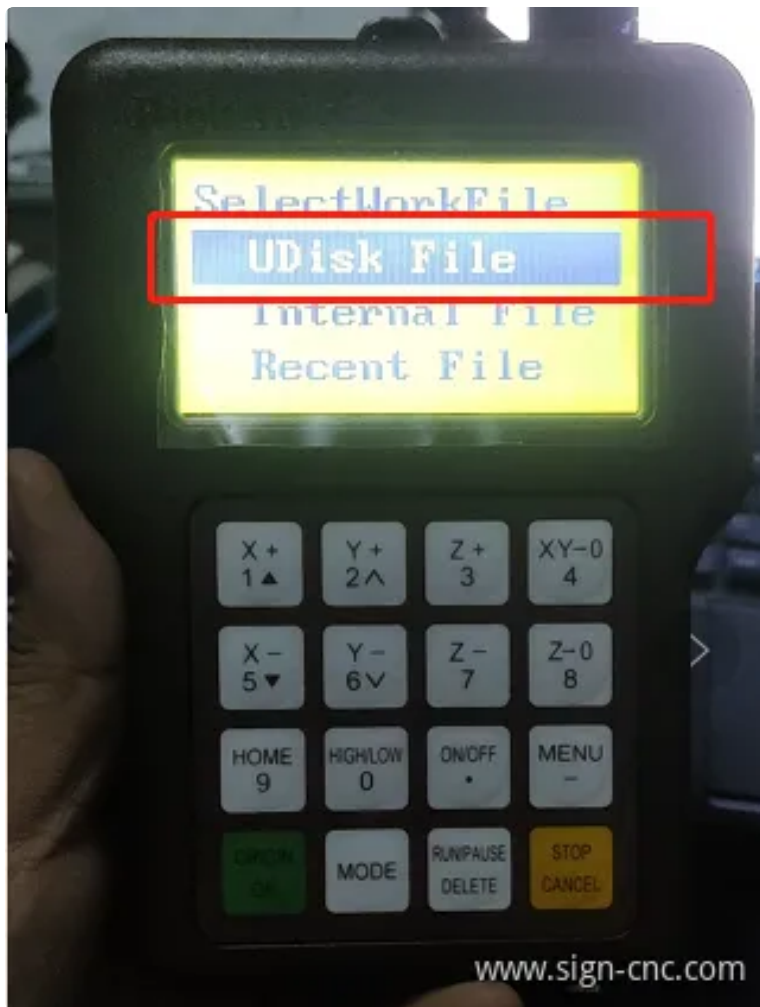
2、 Press "ON/OFF" + "Z+" ,raise the spindle gear .



3、 Press "ON/OFF" + "Z-" ,Lower the spindle gear.

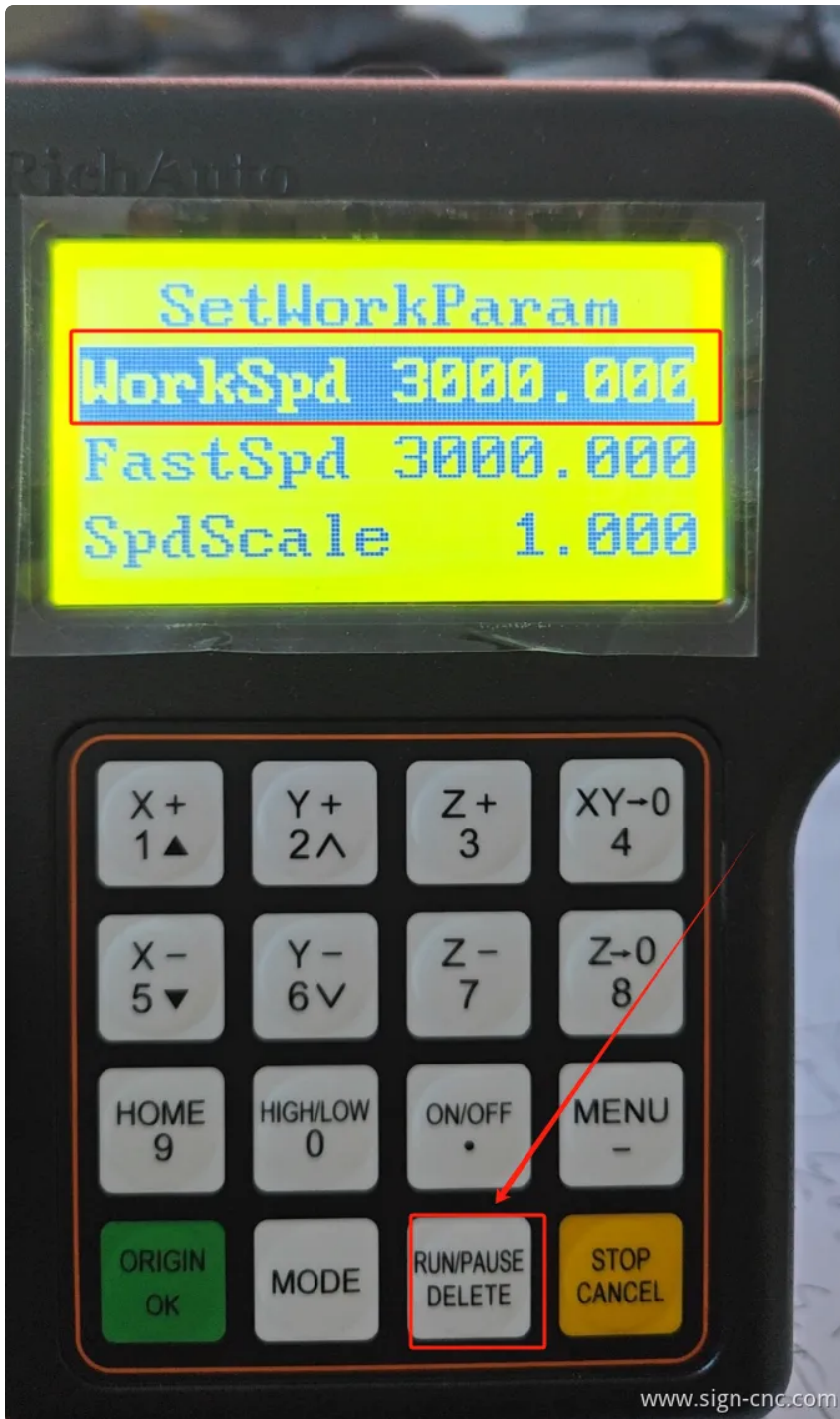
Run the processing program

1. Set the workpiece origin
2. Click the "DELETE" button.
3. Select a file system, as shown in the following figure:





4.set work param,and press "delete" to change.



5. After selecting the file, click OK"

suspend or cancel processing

1. When the program is running automatically, press the "RUN/PAUSE" button to PAUSE the processing.
2. When the program runs automatically, press the "STOP/CANCEL" button to CANCEL the processing.

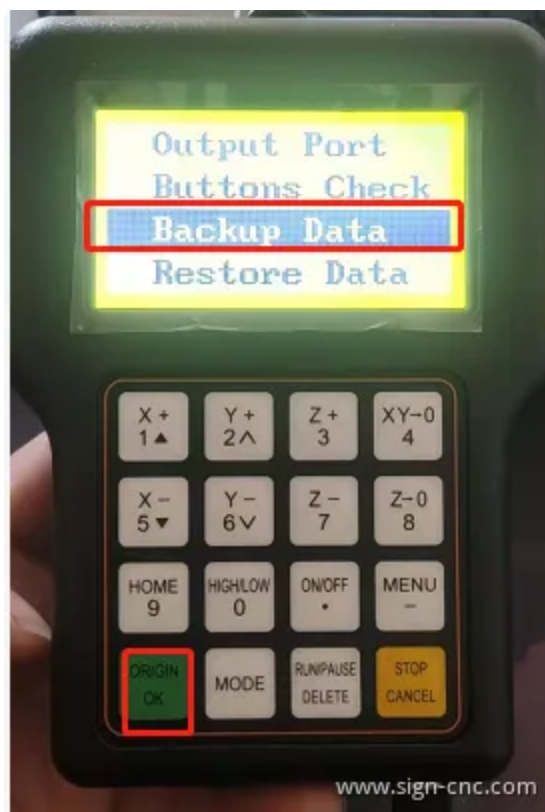
Emergency stop

in all states, press the "Estop" button to stop the machine operation urgently.

Backup and restoration

backup parameters



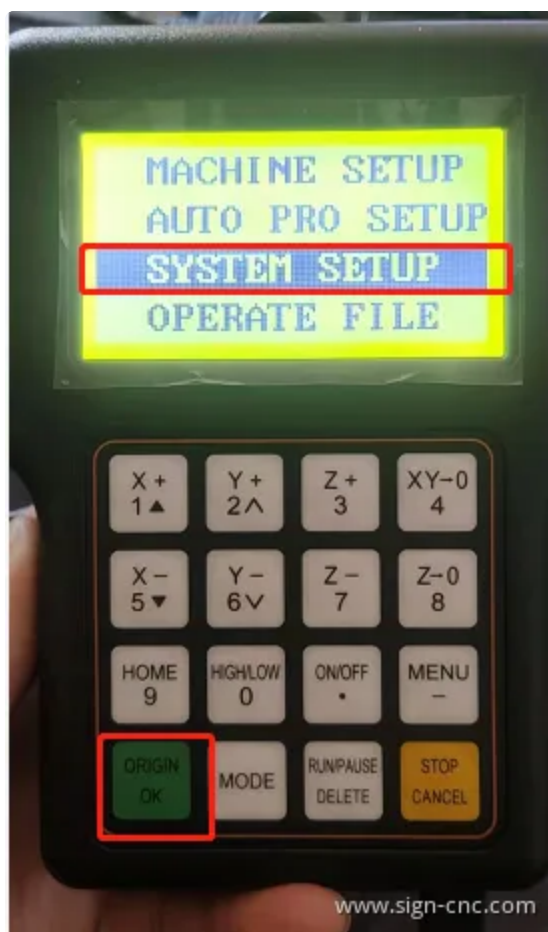


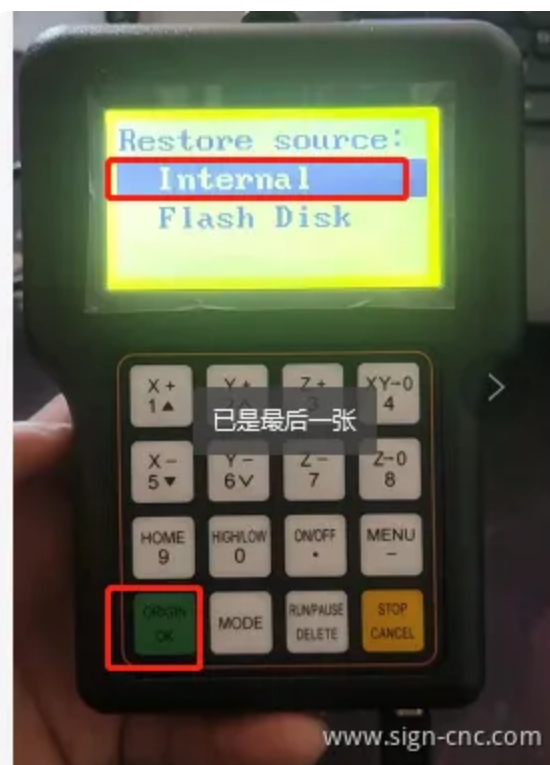
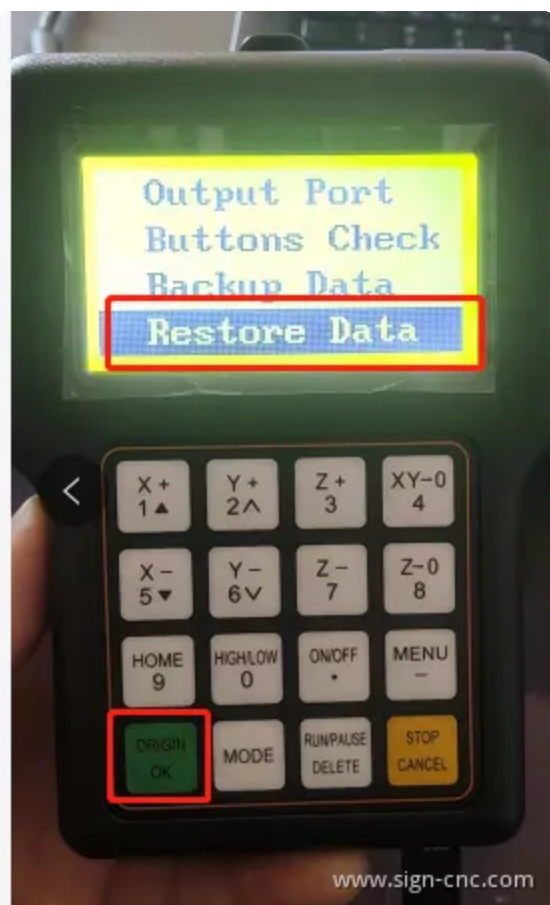


the preceding figure shows an Internal Backup. You need to insert a USB flash disk to the backup. After the backup, a data.bak file is generated in the root directory of the USB flash disk. The file is a backup file.

Recovery parameters







the preceding figure shows the internal recovery parameters. If you need the recovery parameters of the U disk, copy the data.bak backup file to the root directory of the U disk and insert it into the handle. Then, perform the recovery.

Sample program

test all functions G code

```
1
2   G90 G49
3   G40 G80
4   G54
5   G53 Z0
6   M03 S24000
7   G0 X0.0000 Y0.0000
8   G0 X110.0000 Y112.0454 Z50.0000
9   G0   Z20.0000
10  G1   Z-20.0000 F3000.0
11  G0   Z50.0000
12  G0 X210.0000
13  G0   Z20.0000
14  G1   Z-20.0000 F3000.0
15  G0   Z50.0000
16  G0 X310.0000
17  G0   Z20.0000
18  G1   Z-20.0000 F3000.0
19  G0   Z50.0000
20  G0 X410.0000
21  G0   Z20.0000
22  G1   Z-20.0000 F3000.0
23  G0   Z50.0000
24  G0 X510.0000
25  G0   Z20.0000
26  G1   Z-20.0000 F3000.0
27  G0   Z50.0000
28  G0 X610.0000
29  G0   Z20.0000
30  G1   Z-20.0000 F3000.0
31  G0   Z50.0000
32  G0 X710.0000
33  G0   Z20.0000
34  G1   Z-20.0000 F3000.0
35  G0   Z50.0000
36  G0 X810.0000
37  G0   Z20.0000
38  G1   Z-20.0000 F3000.0
39  G0   Z50.0000
40  G0 X910.0000
41  G0   Z20.0000
42  G1   Z-20.0000 F3000.0
43  G0   Z50.0000
44  G0 X1010.0000
45  G0   Z20.0000
```

46	G1	Z-20.0000	F3000.0
47	G0	Z50.0000	
48	G0	X1110.0000	
49	G0	Z20.0000	
50	G1	Z-20.0000	F3000.0
51	G0	Z50.0000	
52	G0	Y212.0454	
53	G0	Z20.0000	
54	G1	Z-20.0000	F3000.0
55	G0	Z50.0000	
56	G0	X1010.0000	
57	G0	Z20.0000	
58	G1	Z-20.0000	F3000.0
59	G0	Z50.0000	
60	G0	X910.0000	
61	G0	Z20.0000	
62	G1	Z-20.0000	F3000.0
63	G0	Z50.0000	
64	G0	X810.0000	
65	G0	Z20.0000	
66	G1	Z-20.0000	F3000.0
67	G0	Z50.0000	
68	G0	X710.0000	
69	G0	Z20.0000	
70	G1	Z-20.0000	F3000.0
71	G0	Z50.0000	
72	G0	X610.0000	
73	G0	Z20.0000	
74	G1	Z-20.0000	F3000.0
75	G0	Z50.0000	
76	G0	X510.0000	
77	G0	Z20.0000	
78	G1	Z-20.0000	F3000.0
79	G0	Z50.0000	
80	G0	X410.0000	
81	G0	Z20.0000	
82	G1	Z-20.0000	F3000.0
83	G0	Z50.0000	
84	G0	X310.0000	
85	G0	Z20.0000	
86	G1	Z-20.0000	F3000.0
87	G0	Z50.0000	
88	G0	X210.0000	
89	G0	Z20.0000	
90	G1	Z-20.0000	F3000.0
91	G0	Z50.0000	
92	G0	X110.0000	
93	G0	Z20.0000	

```

94      G1      Z-20.0000 F3000.0
95      G0      Z50.0000
96      G0      Y412.0454
97      G1      Z-10.0000 F3000.0
98      G0      Z50.0000
99      G0      X0.0000 Y0.0000
100
101      M5
102      G53 Z0
103      M30
104

```

A11 User Manual

attachment System User Manual

Maintenance and maintenance

Spindle part

Water-cooled spindle

Before using the water-cooled spindle, ensure that the cooling water pump (Chiller) is connected and circulates normally. After shutdown, the circulating water in the main shaft must be discharged, especially in winter, all the cooling water must be discharged, otherwise the main shaft may be frozen due to water freezing.

Not used for a long time, clean the spindle, and remove the tool and knife clip from the main shaft, brush oil, to prevent the main shaft from rusting.

Air-cooled spindle

Not used for a long time, and remove the tool and knife clip from the main shaft, brush oil, to prevent the main shaft from rusting.

Lubrication part

oil jug is filled with 32# guide rail lubricating oil (or lubricating oil with similar viscosity, which can be selected in winter viscosity

<https://fvh9y7.yuque.com/sign-cnc/tpvpye/fupszd?singleDoc#> 《Manual for lubrication system》